

Analytics Report

1. Introduction

The **Enterprise IoT & Embedded Automation Platform** collects real-time data from multiple sensors connected to an ESP32 controller. The collected parameters include temperature, humidity, ambient light, motion, and machine-value data. These parameters are transmitted to the Blynk IoT platform for monitoring, visualization, and analysis.

The analytics component is used to identify abnormal operating conditions, monitor system performance, support automated decision-making, and provide early indications of potential equipment or environmental problems.

2. Data Parameters Analyzed

Parameter	Source	Purpose
Temperature	DHT22	Monitor environmental/operating temperature
Humidity	DHT22	Monitor environmental humidity
Ambient Light	LDR	Determine lighting conditions
Motion	PIR Sensor	Detect human/activity presence
Machine Value	Potentiometer	Simulate machine operating condition
Relay Status	Relay	Monitor appliance/load control
Load Status	Simulated Load	Verify load operation
Operation Mode	Blynk	Identify AUTO/MANUAL operation
Alert Level	ESP32 Logic	Identify normal, warning, and critical conditions
Emergency Status	ESP32 Logic	Indicate critical system conditions

3. Data Collection and Monitoring

Sensor values are periodically read by the ESP32 and processed according to the programmed control logic. The processed values are sent to Blynk using virtual datastreams.

The system continuously monitors:

- Environmental conditions
- Machine operating value
- Motion/activity
- Relay and load state
- Operating mode
- Alert conditions

- Emergency status

The Blynk dashboard provides a centralized interface for viewing current values and historical trends.

4. Threshold-Based Analytics

Threshold-based analysis is implemented to identify abnormal conditions.

Parameter	Threshold	Interpretation
Ambient Light	2000	Used for automatic lighting control
Machine Value	> 3000	Warning condition
Machine Value	> 3500	Critical condition
Temperature	$\geq 35^{\circ}\text{C}$	Warning condition
Temperature	$\geq 40^{\circ}\text{C}$	Critical condition
Motion	Detected	Activity/warning condition

The threshold mechanism allows the system to classify operating conditions without requiring complex machine-learning algorithms.

5. Alert-Level Analysis

The system uses three logical operating conditions.

5.1 Normal Condition

The system remains in the normal state when temperature and machine-value readings remain within their defined operating limits.

- Alert Level = 0
- Emergency Status = 0
- Alert Message = SYSTEM NORMAL

5.2 Warning Condition

A warning is generated when one or more monitored parameters exceed their warning thresholds.

Examples include:

- Temperature $\geq 35^{\circ}\text{C}$
- Machine Value > 3000
- Motion detected

During this condition:

- Alert Level = 1

- Emergency Status = 0

5.3 Critical Condition

A critical condition is generated when the machine value or temperature reaches the critical threshold.

Examples include:

- Machine Value > 3500
- Temperature $\geq 40^{\circ}\text{C}$

During a critical condition:

- Alert Level = 2
- Emergency Status = 1

This classification helps operators distinguish between normal operation, conditions requiring attention, and potentially dangerous conditions.

6. Environmental Data Analysis

6.1 Temperature Analysis

Temperature is monitored continuously through the DHT22 sensor.

The analytics logic categorizes temperature conditions as:

- **Below 35°C :** Normal
- **35°C to below 40°C :** Warning
- **40°C or above:** Critical

Increasing temperature can indicate an abnormal environmental condition or excessive operating stress. Therefore, temperature monitoring can be used as an early indicator for preventive maintenance in a real industrial implementation.

6.2 Humidity Analysis

Humidity is monitored along with temperature to provide additional environmental information.

Humidity trends can be observed through the Blynk dashboard and historical chart. Although the current implementation does not trigger a specific emergency condition based on humidity, the collected data can be used for future environmental analysis and threshold configuration.

7. Machine-Value Analysis

The machine-value input is simulated using a potentiometer and represents a variable machine operating parameter.

The system analyzes the machine value using two thresholds:

- **Normal:** Machine Value ≤ 3000

- **Warning:** Machine Value > 3000
- **Critical:** Machine Value > 3500

As the simulated machine value increases, the system changes the alert level accordingly.

This demonstrates how an industrial IoT system can monitor equipment parameters and identify abnormal operating conditions.

8. Motion and Lighting Analysis

The PIR sensor is used to identify motion or activity, while the LDR measures ambient light.

In **AUTO mode**, the relay control follows this logic:

If ambient light is below the configured threshold AND motion is detected → Relay ON

Otherwise:

Relay OFF

Light Condition	Motion	Relay
Dark	Detected	ON
Dark	Not detected	OFF
Bright	Detected	OFF
Bright	Not detected	OFF

This demonstrates sensor-based automation where multiple sensor conditions are combined to make a control decision.

9. Relay and Load Performance Analysis

The relay acts as the interface between the ESP32 control system and the simulated load.

The system can operate the relay in two modes:

AUTO Mode

The relay is controlled automatically according to light and motion conditions.

MANUAL Mode

The user can control the relay using the Blynk manual control interface.

This dual-mode architecture provides flexibility between automatic operation and operator-controlled operation.

10. Historical Data Analysis

The Blynk dashboard provides charts for selected parameters such as:

- Temperature
- Humidity
- Motion

Historical data can be examined to identify:

- Increasing or decreasing temperature trends
- Changes in humidity
- Repeated motion activity
- Abnormal operating patterns
- System behavior over time

Historical trends help operators understand system behavior rather than relying only on instantaneous sensor readings.

11. Predictive Maintenance Perspective

Although the current project uses **threshold-based analytics rather than machine-learning predictive models**, the collected machine and environmental data provides a foundation for future predictive maintenance.

Repeated occurrences of:

- High machine values
- Increasing temperature
- Frequent warning conditions
- Critical alerts

could indicate potential equipment stress or abnormal operation.

In a future implementation, historical sensor data could be processed using statistical analysis or machine-learning algorithms to predict failures before they occur.

12. Energy-Efficiency Analysis

The automation logic contributes to energy efficiency by operating the simulated load only when required.

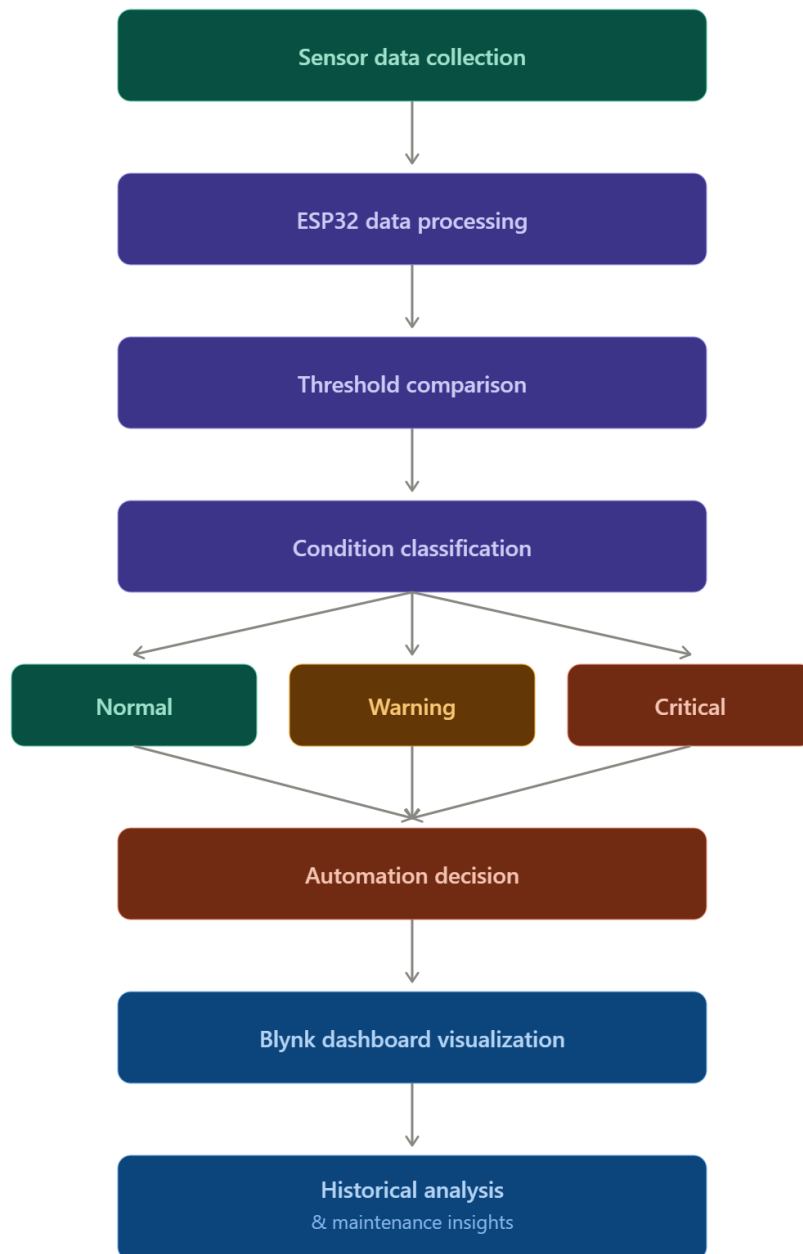
In AUTO mode, the relay is activated only when:

1. The ambient light level indicates insufficient lighting, and
2. Motion is detected.

This prevents the load from remaining continuously ON when the area is sufficiently illuminated or unoccupied.

13. Analytics Workflow

The overall analytics process can be represented as:



This workflow integrates embedded processing, cloud monitoring, automation, and analytics into a single IoT platform.

14. Key Analytical Findings

Based on the implemented simulation and testing, the following observations were made:

1. The ESP32 successfully collects data from multiple sensors.
2. Temperature and humidity values can be monitored continuously.
3. Ambient light and motion data can be combined for automated lighting control.
4. Machine-value variations can be used to simulate equipment operating conditions.
5. Threshold-based analysis successfully identifies warning and critical conditions.
6. Relay operation responds to the configured automation logic.
7. AUTO and MANUAL modes provide flexible control.
8. Blynk provides centralized real-time monitoring.
9. Historical charts can be used to identify trends in monitored parameters.
10. The collected data can serve as a foundation for future predictive-maintenance analytics.

15. Limitations of the Current Analytics

The current implementation has some limitations:

- Machine data is simulated using a potentiometer rather than a real industrial sensor.
- Predictive maintenance is demonstrated conceptually through threshold monitoring.
- No machine-learning model is implemented.
- The Wokwi environment simulates rather than physically measures industrial conditions.
- Long-term historical datasets are not available in the prototype.
- Advanced statistical anomaly detection is not implemented.

These limitations are acceptable for the prototype stage and can be addressed in a future physical deployment.

16. Future Analytics Enhancements

The analytics module can be further enhanced by:

- Integrating real industrial sensors.
- Storing long-term sensor data in a database.
- Implementing machine-learning-based anomaly detection.
- Developing predictive-maintenance models.
- Developing automated analytics reports.

- Implementing advanced statistical dashboards.
- Implementing trend-based alerts.
- Analyzing energy consumption.
- Using historical machine data to estimate failure probability.
- Integrating additional industrial communication protocols.

17. Conclusion

The analytics component of the **Enterprise IoT & Embedded Automation Platform** demonstrates how sensor data can be transformed into meaningful operational information. The system combines real-time monitoring, threshold-based analysis, automated control, alert classification, and historical visualization.

The implementation provides a foundation for more advanced industrial IoT analytics and predictive maintenance. With the integration of real industrial sensors, larger datasets, and machine-learning techniques, the prototype can be extended into a comprehensive enterprise monitoring and predictive-maintenance solution.